

Recap Lecture 16

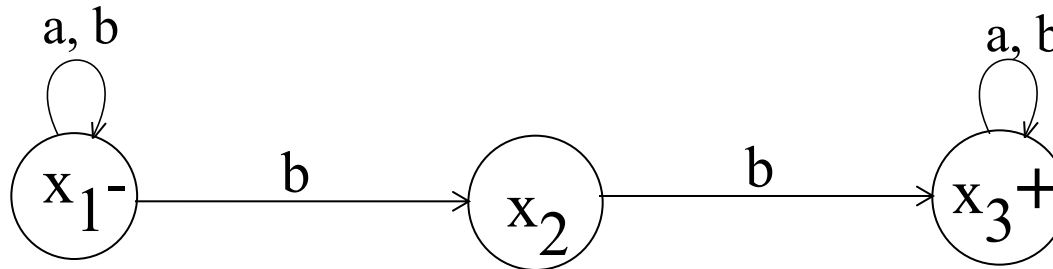
- Examples of Kleene's theorem part III (method 3), NFA, examples, avoiding loop using NFA, example, converting FA to NFA, examples, applying an NFA on an example of maze, NFA with null string, examples, RE corresponding to NFA with null string (task), converting NFA to FA (method 1,2,3) examples, NFA and Kleene's theorem method 1, examples, NFA and Kleene's theorem method 2, examples

NFA to FA continued ...

Method 3: As discussed earlier that in an NFA, there may be more than one transition for a certain letter and there may not be any transition for certain letter, so starting from the initial state corresponding to the initial state of given NFA, the transition table along with new labels of states, of the corresponding FA, can be built introducing an empty state for a letter having no transition at certain state and a state corresponding to the combination of states, for a letter having more than one transitions. Following are the examples

Example

- Consider the following NFA which accepts the language of strings **containing bb**



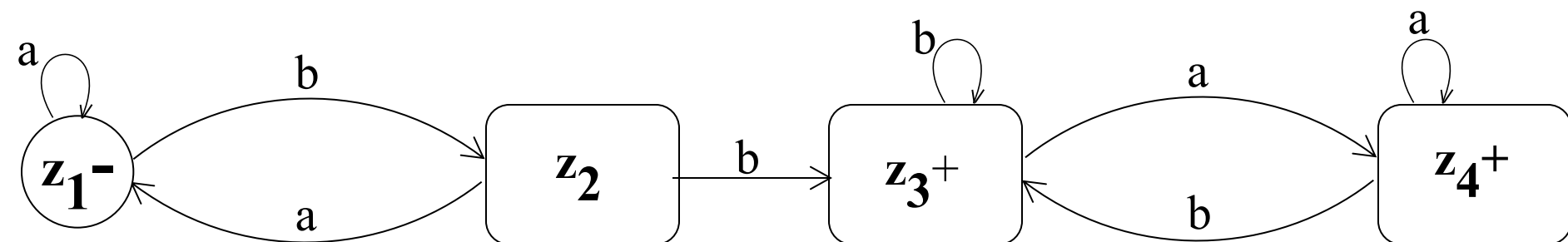
Using the method discussed earlier, the transition table corresponding to the required FA may be constructed as

Example continued ...

Old States	New States after reading	
	a	b
$z_1 \sqcap x_1$	$x_1 \sqcap z_1$	$(x_1, x_2) \sqcap z_2$
$z_2 \sqcap (x_1, x_2)$	$(x_1, \sqcap) \sqcap x_1 \sqcap z_1$	$(x_1, x_2, x_3) \sqcap z_3$
$z_3 + \sqcap (x_1, x_2, x_3)$	$(x_1, x_3) \sqcap z_4$	$(x_1, x_2, x_3) \sqcap z_3$
$z_4 + \sqcap (x_1, x_3)$	$(x_1, x_3) \sqcap z_4$	$(x_1, x_2, x_3) \sqcap z_3$

The corresponding transition diagram follows as₄

Example continued ...



NFA and Kleene's Theorem

It has been discussed that, by Kleene's theorem part III, there exists an FA corresponding to a given RE. If the given RE is as simple as $r=aa+bbb$ or $r=a(a+b)^*$, the corresponding FAs can easily be constructed. However, for a complicated RE, the RE can be decomposed into simple REs corresponding to which the FAs can easily be constructed and hence, using the method, constructing the FAs corresponding to sum, concatenation and closure of FAs, the required FA can also be constructed. It is to be noted that NFAs also help in proving Kleene's theorem part III, as well. Two methods are discussed in the following.

NFA and Kleene's Theorem

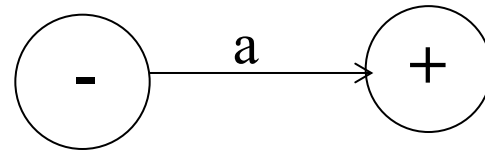
Method 1: The method is discussed considering the following example.

Example: To construct the FAs for the languages $L_1 = \{a\}$, $L_2 = \{b\}$ and $L_3 = \{?\}$

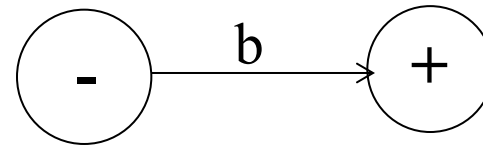
Step 1: Build NFA_1 , NFA_2 and NFA_3 corresponding to L_1 , L_2 and L_3 , respectively as shown in the following diagram

NFA and Kleene's Theorem (method 1) continued ...

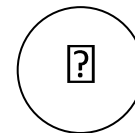
- NFA_1



- NFA_2



- NFA_3

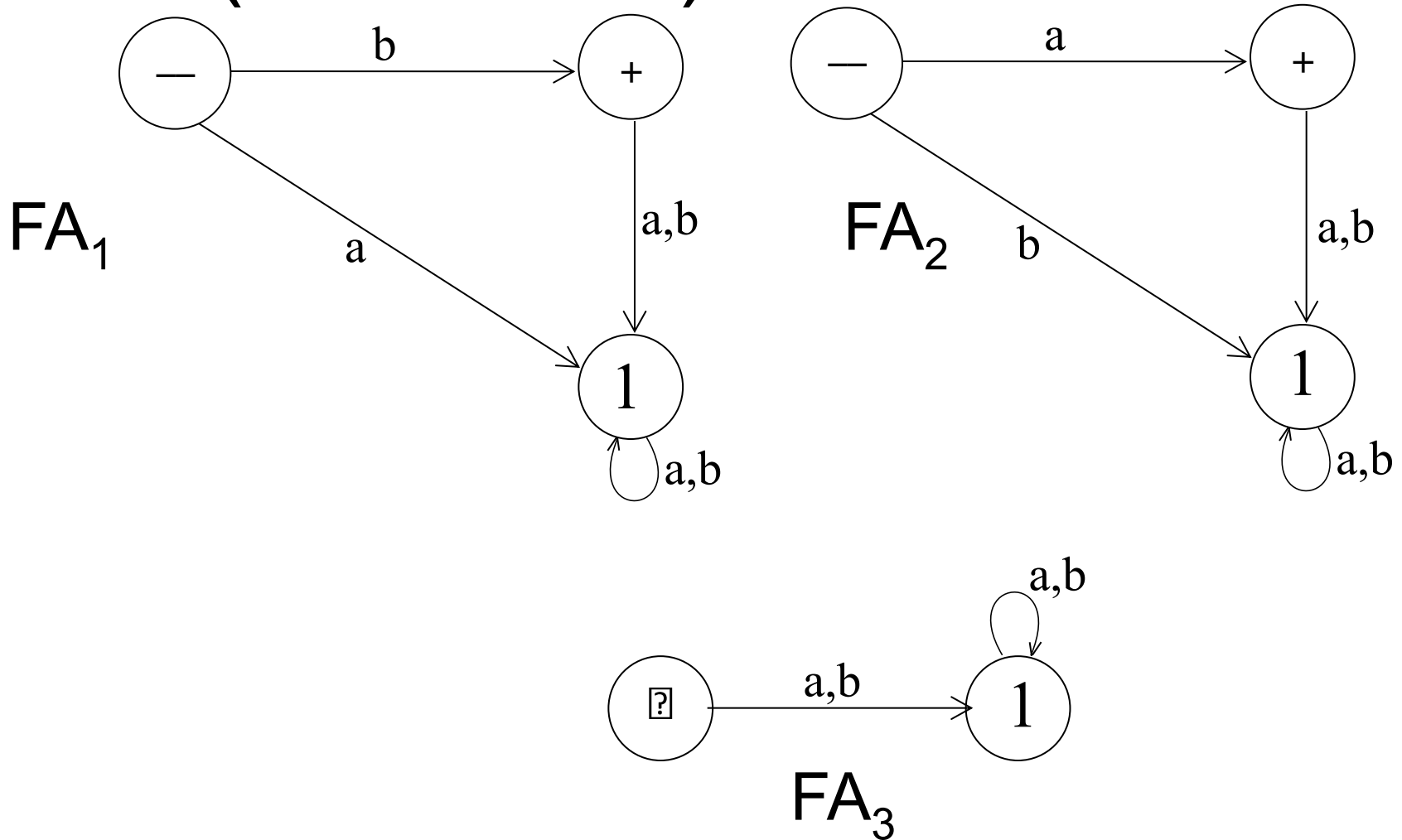


NFA and Kleene's Theorem (method 1) continued ...

Step 2:

As discussed earlier for every NFA there is an FA equivalent to it, hence there must be FAs for the above mentioned NFAs as well. The corresponding FAs can be considered as follows

NFA and Kleene's Theorem (method 1) continued ...



NFA and Kleene's Theorem

method 2

It may be observed that if an NFA can be built corresponding to union, concatenation and closure of FAs corresponding to the REs, then converting the NFA, thus, obtained into an equivalent FA, this FA will correspond to the given RE.

Followings are the procedures showing how to obtain NFAs equivalent to union, concatenation and closure of FAs

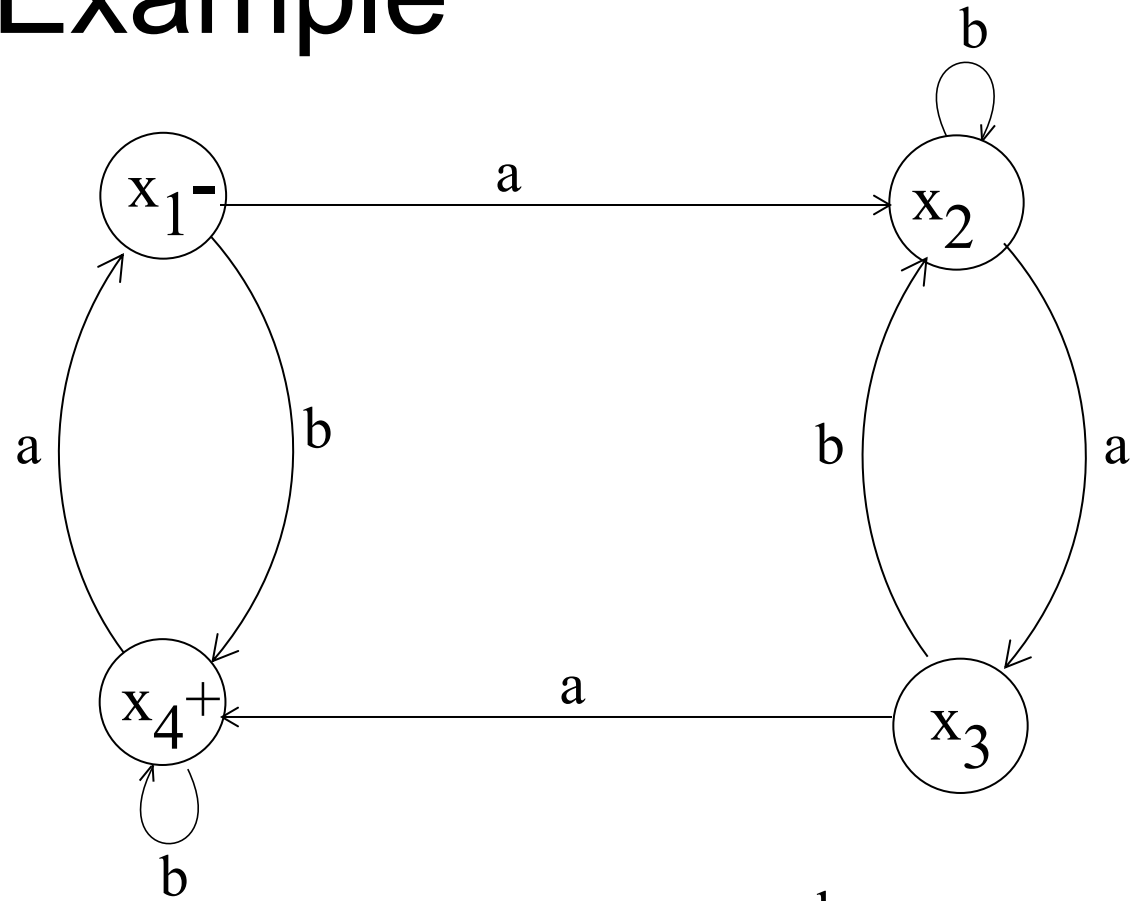
NFA corresponding to Union of FAs

Method:

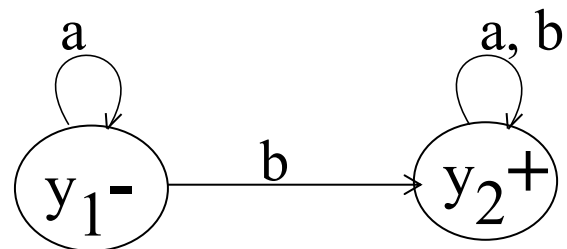
Introduce a new start state and connect it with the states originally connected with the old start state with the same transitions as the old start state, then remove the –ve sign of old start state. This creates non-determinism and hence results in an NFA.

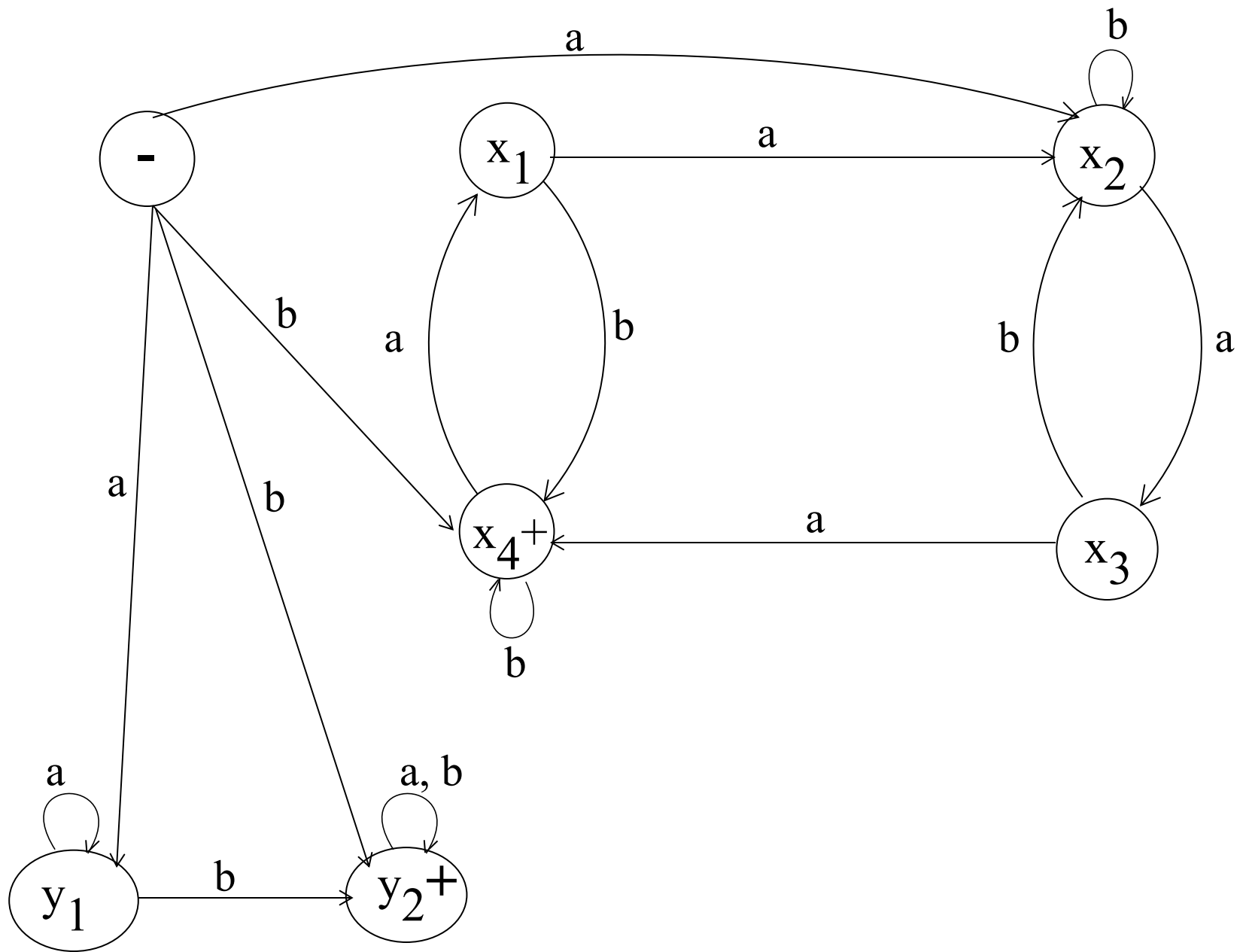
Example

- FA_1



- FA_2





NFA equivalent to $FA_1 U FA_2$

Summing Up

- converting NFA to FA (method 3), example, NFA and Kleene's theorem method 1, examples, NFA and Kleene's theorem method 2 , NFA corresponding to union of FAs, example